1. Logit transform

$$\text{logit}(AUC) = \ln \frac{AUC}{1 - AUC}$$

$$SE_{\text{logit}} = \frac{SE_{AUC}}{AUC\,(1 - AUC)}$$

scripts/05_meta_analysis.py — logit(), delta method

2. Hanley–McNeil standard error

$$Q_1 = \frac{AUC}{2 - AUC}, \quad Q_2 = \frac{2\,AUC^2}{1 + AUC}$$

$$SE(AUC) = \sqrt{\frac{AUC(1-AUC) + (n_1-1)(Q_1-AUC^2) + (n_2-1)(Q_2-AUC^2)}{n_1 n_2}}$$

scripts/05_meta_analysis.py — se_hanley_mcneil(); Hanley & McNeil, 1982

3. Paule–Mandel random effects, modified Hartung–Knapp CI

$$\sum_i w_i^*(y_i - \hat\theta)^2 = k - 1, \quad w_i^* = \frac{1}{v_i + \hat\tau^2}, \quad \hat\theta = \frac{\sum w_i^* y_i}{\sum w_i^*}$$

$$SE_{mHK} = \max\left(\sqrt{\frac{\sum w_i^*(y_i - \hat\theta)^2}{(k-1)\sum w_i^*}},\ \frac{1}{\sqrt{\sum w_i^*}}\right), \quad \hat\theta \pm t_{k-1}\,SE_{mHK}$$

scripts/05_meta_analysis.py — paule_mandel_tau2(), pool_re()
Paule & Mandel 1982; Hartung & Knapp 2001; IntHout et al. 2014

4. Egger's regression test

$$\frac{y_i}{SE_i} = a + b\left(\frac{1}{SE_i}\right), \quad t = \frac{\hat a}{SE(\hat a)} \text{ on } k - 2 \text{ df}$$

the intercept a is tested; $a \neq 0$ read as small-study asymmetry

scripts/05_meta_analysis.py — egger_test(); Egger et al., 1997

5. Bivariate model (Reitsma)

$$\text{logit}(Se_i),\ \text{logit}(Sp_i)\ \sim\ \text{BVN}(\mu_{Se}, \mu_{Sp}, \tau^2_{Se}, \tau^2_{Sp}, \rho)$$

joint normal in Se and Sp, correlated by ρ across studies

scripts/15_bivariate_srocc.py; Reitsma et al., 2005

6. Standard error from a reported 95% CI

$$SE = \frac{\text{upper} - \text{lower}}{2 \times 1.959964}$$

scripts/05_meta_analysis.py — se_from_ci()